



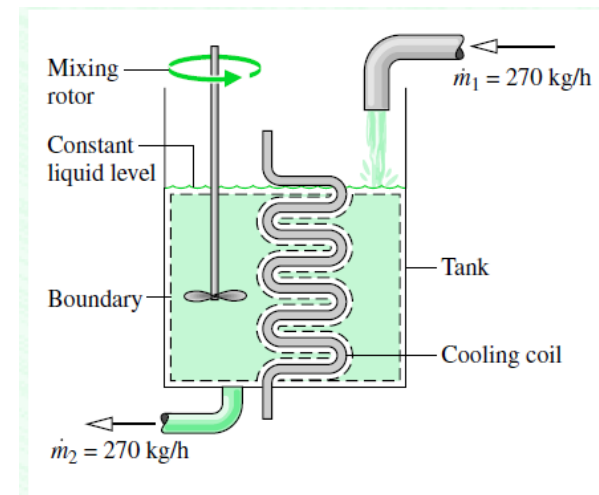
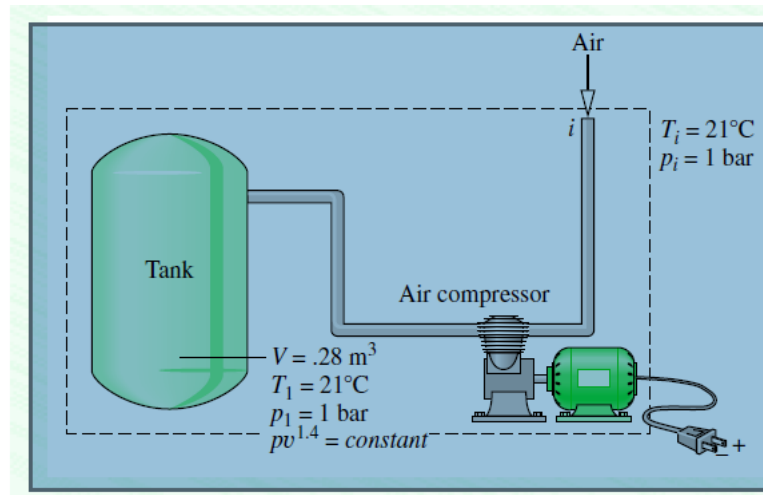
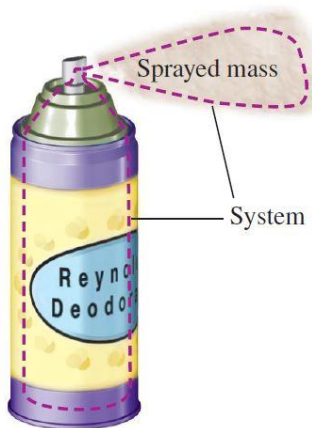
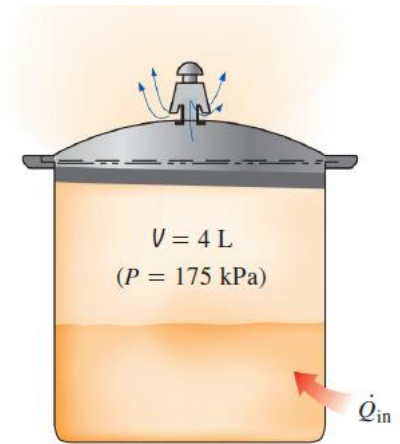
Module-4: First Law of Thermodynamics

Thermodynamics (NMEC103) : (Winter 2024-25)

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Transient Analysis

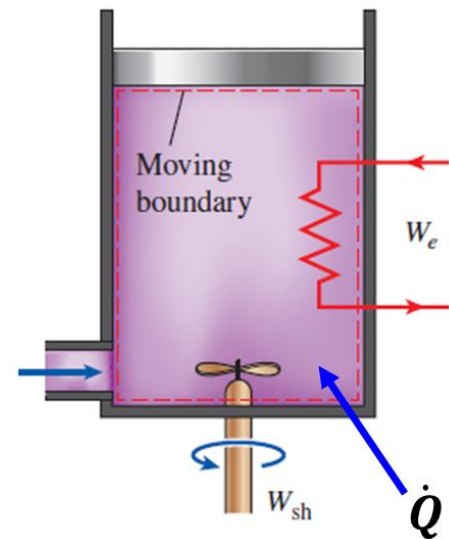
Many devices undergoes **periods of transient (time dependent)** operation in which the **state changes with time**. Examples includes **start up or short down of turbines compressors, and motors**. **Filling and discharging of gas tanks, inflating tires or balloons, cooking with an ordinary pressure cookers**. The property values, work and heat transfer rates and mass transfer rates vary with time during the transient periods, the steady state assumptions are not valid



Unlike Steady flow processes the unsteady flow processes start and end over a finite period of time **instead of continuing indefinitely**. Therefore here we will deal with changes that occur over some period of time interval Δt **(integral form)** instead of **with the rate of changes (change for unit time (differential version))**

An unsteady-flow system, **in some respect**, is similar to closed **system**, except the mass within the system boundaries does not remain constant during a process

An Important difference between steady and unsteady flow systems is that **steady flow systems are fixed in space, size and shape**. Unsteady flow systems are not. **They are usually stationary; that is they are fixed in space, but they may involve moving boundary and thus boundary work**. Unsteady flow system may involve **boundary work**, **shaft work**, **electric work** and **heat interaction also**.



Transient Analysis:

Transient problem arises when the control volume is not in the steady state. Heat, work and mass transfer rates as well as properties at the open boundaries may vary with time. As a result, properties within the control volume vary.

The general energy equation for a control volume is given as:

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_{in} \dot{m}_i \left(h + \frac{v^2}{2} + gz \right)_i - \sum_{out} \dot{m}_e \left(h + \frac{v^2}{2} + gz \right)_e$$

Since some of the terms in the above equation vary with time, the equation must be integrated over time "t" to get the total energy change of the control volume during this interval.

For a control volume which remain fixed in the space so that

$$E_{cv} = U_{cv}$$

$$\int_0^t \frac{dU_{cv}}{dt} dt = \int_0^t \dot{Q}_{cv} dt - \int_0^t \dot{W}_{cv} dt + \sum_{in} \int_0^t \dot{m}_i \left(h + \frac{V^2}{2} + gz \right)_i dt - \sum_{out} \int_0^t \dot{m}_e \left(h + \frac{V^2}{2} + gz \right)_e dt$$

To carry out the above integration, we need to know, how the various properties and rates vary with time.

The integration of the control volume energy term on the left hand side may take any one of the several forms, depending on the type of process under investigation. Among the possibilities are:

$$\int_0^t \frac{dU_{cv}}{dt} dt = \int_0^t dU_{cv} = \int_0^t d(mu)_{cv} = (mu)_{at\ time=t} - (mu)_{at\ time=0} = \Delta U_{cv}$$

$$\int_0^t \frac{dU_{cv}}{dt} dt = \int_0^t dU_{cv} = \int_0^t d(mu)_{cv} = \int (mdu + udm)$$

For a differential change of state for a control volume over a time dt .
Thus

$$\delta Q - \delta W + \sum_{in} \left(h + \frac{V^2}{2} + gz \right) dm_i - \sum_{out} \left(h + \frac{V^2}{2} + gz \right) dm_e = dU_{CV}$$

Integrating over a time period of interest yields

$$Q - W + \int \left(h + \frac{V^2}{2} + dz \right) dm_i - \int \left(h + \frac{V^2}{2} + dz \right) dm_e = \Delta U_{CV}$$

In the analysis of transient flow systems several models are frequently used

Models of Uniform State: A uniform state requires that the state within the control volume at any instant be uniform throughout the control volume or over several regions of the control volume. However, the state within the control volume may change with time. The condition of uniform state implies rapid or instantaneous approach to equilibrium at all time for the mass within the control volume

Model of Uniform Flow: For all time the properties of fluid entering/leaving C.V. at the boundary is uniform and steady. However, the mass flow rate across the particular control surface may vary with time. The condition of uniform flow is frequently met when the flow into a transient system is supplied from a very large reservoir of matter. If they at all vary they are averaged and treated as constant for the entire process

Continuity Equation:

$$\sum_{in} \dot{m} - \sum_{out} \dot{m} = \frac{dm_{CV}}{dt}$$

If mass flow rate varies

$$\sum_{in} \int_0^t \dot{m}_i dt - \sum_{out} \int_0^t \dot{m}_e dt = \Delta m_{CV}$$



The condition of uniform flow is frequently met when the flow in to a transient system is supplied from a very large reservoir of matter.



For the time period “t” the continuity equation

$$(m_f - m_i)_{cv} = \sum m_e - \sum m_i$$

$$\frac{dE_{CV}}{dt} = \dot{Q}_{CV} - \dot{W}_{CV} + \sum_{in} \dot{m}_i \left(h + \frac{V^2}{2} + gz \right)_i - \sum_{out} \dot{m}_e \left(h + \frac{V^2}{2} + gz \right)_e$$

$$\int_0^t \frac{dE_{CV}}{dt} dt = \int_0^t \dot{Q}_{CV} dt - \int_0^t \dot{W}_{CV} dt + \sum_{in} \int_0^t \dot{m}_i \left(h + \frac{V^2}{2} + gz \right)_i dt - \sum_{out} \int_0^t \dot{m}_e \left(h + \frac{V^2}{2} + gz \right)_e dt$$

$$\int_0^t \dot{Q}_{CV} dt = Q_{CV}$$

$$\sum_{in} \int_0^t \dot{m}_i \left(h + \frac{V^2}{2} + gz \right)_i dt = \sum m_i \left(h + \frac{V^2}{2} + gz \right)$$

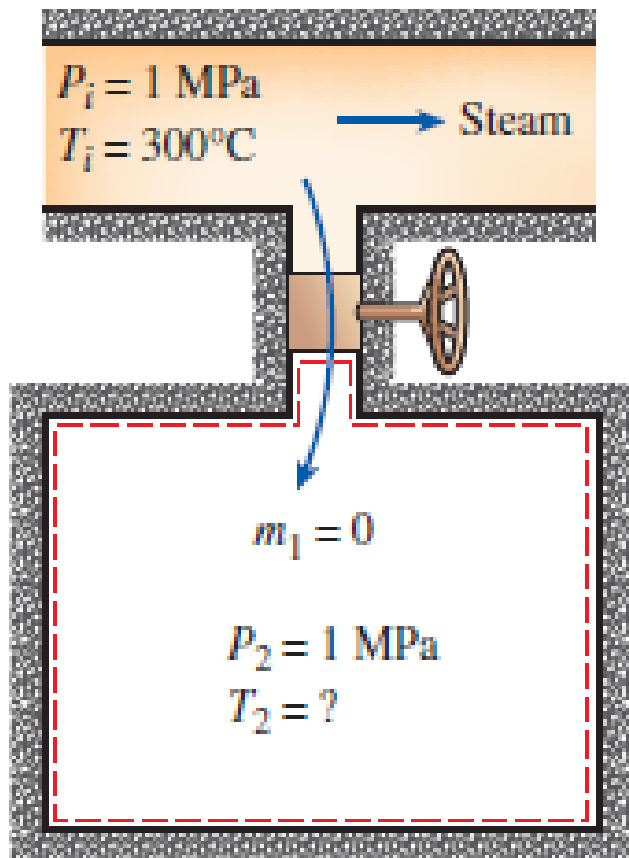
$$\int_0^t \dot{W}_{CV} dt = W_{CV}$$

$$\sum_{out} \int_0^t \dot{m}_e \left(h + \frac{V^2}{2} + gz \right)_e dt = \sum m_e \left(h + \frac{V^2}{2} + gz \right)$$

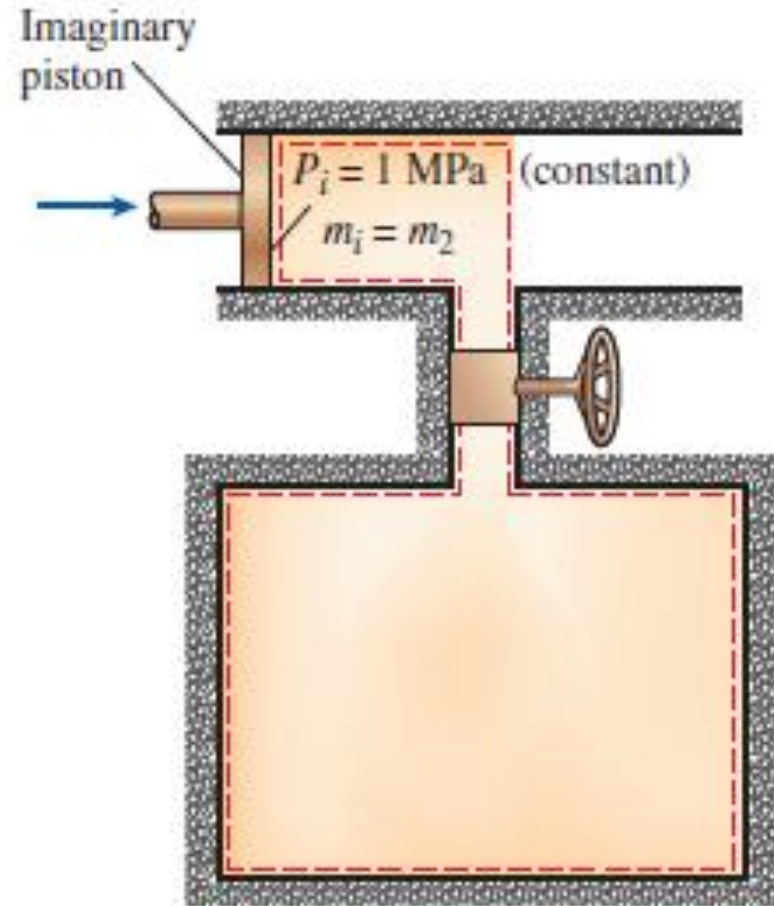
$$\int_0^t \frac{dE_{CV}}{dt} dt = \int_0^t \frac{d}{dt} \left[m \left(u + \frac{V^2}{2} + gz \right) \right]_{cv} dt = \left[m_f \left(u + \frac{V^2}{2} + gz \right)_f - m_i \left(u + \frac{V^2}{2} + gz \right)_i \right]$$

Charging Process:

Consider an **evacuated rigid tank** connected through a **closed valve** to a **high pressure line**. The problem can be analysed using a **control volume approach** or by using a **control mass approach**.



(a) Flow of steam into an evacuated tank



(b) The closed-system equivalence

Control Volume Approach:

A control volume has been selected as shown in fig.1 (indicated by the red dotted line). We shall assume that the volume in the region of the valve and entrance pipe is negligible.

$$Q - W + \int_i \left(h + \frac{V^2}{2} + dz \right) dm_i - \int_e \left(h + \frac{V^2}{2} + dz \right) dm_e = \Delta U_{CV}$$

The net work term is zero (rigid tank). Usually the kinetic and potential energies of any unit mass in the line may be neglected. Thus the conservation of energy equation for the selected control volume reduces to

$$Q + \int h dm = \Delta U_{CV}$$

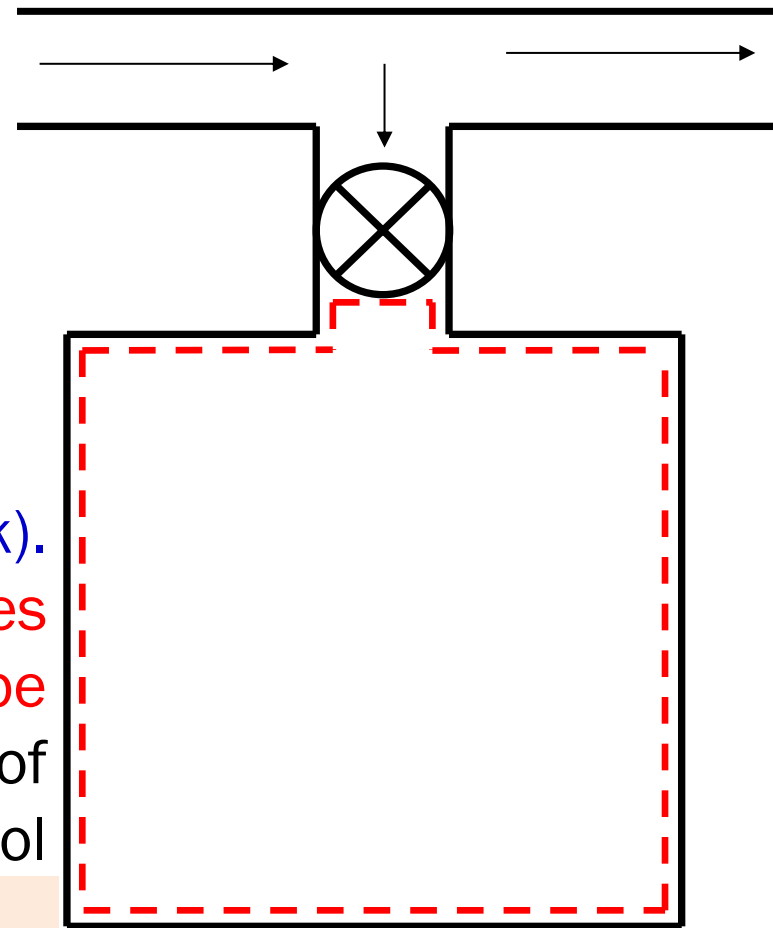


Fig.1. Open System

If the tank is **heavily insulated** or the tank were allowed to fill rapidly , the amount of heat transfer will be quite small even if an appreciable temperature gradient existed between the contents of tank and the environment. Therefore we will analyse process on the basis of negligible heat transfer.

$$Q + \int h dm = \Delta U_{CV}$$

$$\int h dm = m_f u_f - m_i u_i$$

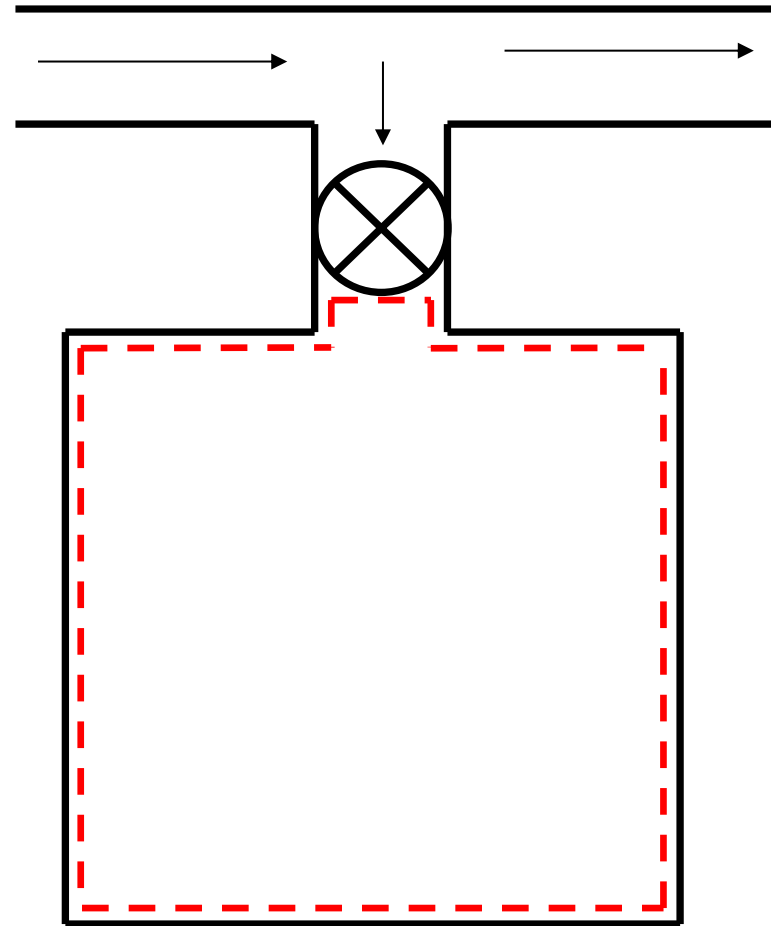


Fig.1. Open System

Frequently the integral of $\int h dm$ is difficult to evaluate, because the enthalpy of the mass crossing the control surface varies w.r.t time. However, in our particular case we can assume that the quantity of fluid bled from the line is small compared to the quantity flowing through the line. Consequently the properties of the fluid at the entrance to the valve remain constant. So we have a case of uniform flow. If we designate the pipeline condition by subscript L

$$h_L(m_L - 0) = m_f u_f - m_i u_i$$

$$\text{If } m_i = 0 \quad h_L m_L = m_L u_f$$

If the gas charged is an ideal gas with constant specific heat:

$$c_p T_L = c_v T_f$$

$$T_f = \frac{c_p}{c_v} T_L = \gamma T_L$$

Note: $T_f > T_L$

For the flow of an ideal gas in to a rigid tank under the following major assumptions as γ for many gases ranges from 1.25 to 1.4 absolute temperature rise may be 40 %

- ❖ Adiabatic, rigid control volume
- ❖ Uniform state within the tank at any instant of time
- ❖ Negligible kinetic energy of the inflowing gas
- ❖ Uniform inlet flow

If the tank were filled with an incompressible fluid, then

$$u_f = CT_f$$

$$h_L = CT_L + p_L v$$

$$T_f = T_L + \frac{p_L v}{C}$$

Control Mass Approach. Now we will address the same problem using control mass approach

Mass Δm entered into the control volume (Figure 1) in time Δt is included in the control mass and shown by the green dotted line in fig. 2

Applying 1st Law to the closed system bounded by the dotted line

$$Q - W = \Delta E$$

Neglecting the KE and PE and assuming adiabatic condition

$$-W = \Delta U = m_f u_f - m_i u_i$$

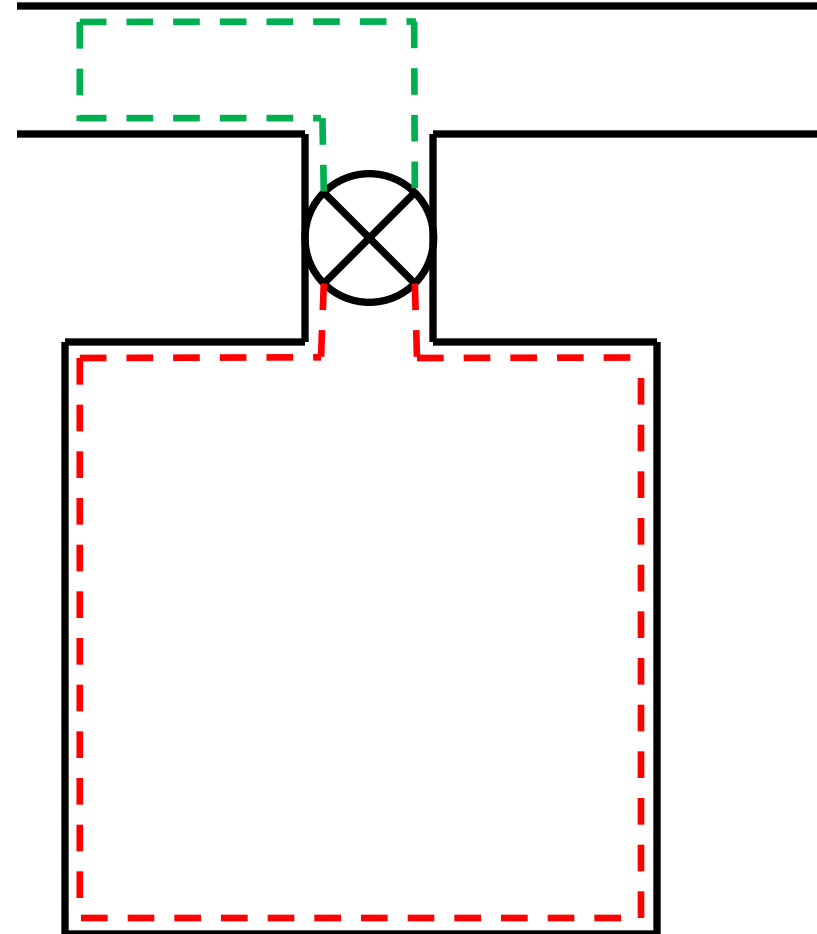


Fig. 2. Closed System

Unlike the control-volume approach however, neither the work term nor the initial mass term is zero in the present approach. The work done is the boundary work related to pushing the mass contained within the system but outside the valve in to the tank

$$-P(V_f - V_i) = m_f u_f - m_i u_i$$

$$-P(0 - m_L v_L) = m_f u_f - m_i u_i$$

$$m_f = m_i = m_L$$

$$u_i = u_L$$

$$m_L(u_L + P v_l) = m_L u_f$$

$$\Rightarrow m_L h_L = m_L u_f$$

$$\Rightarrow h_L = u_f$$

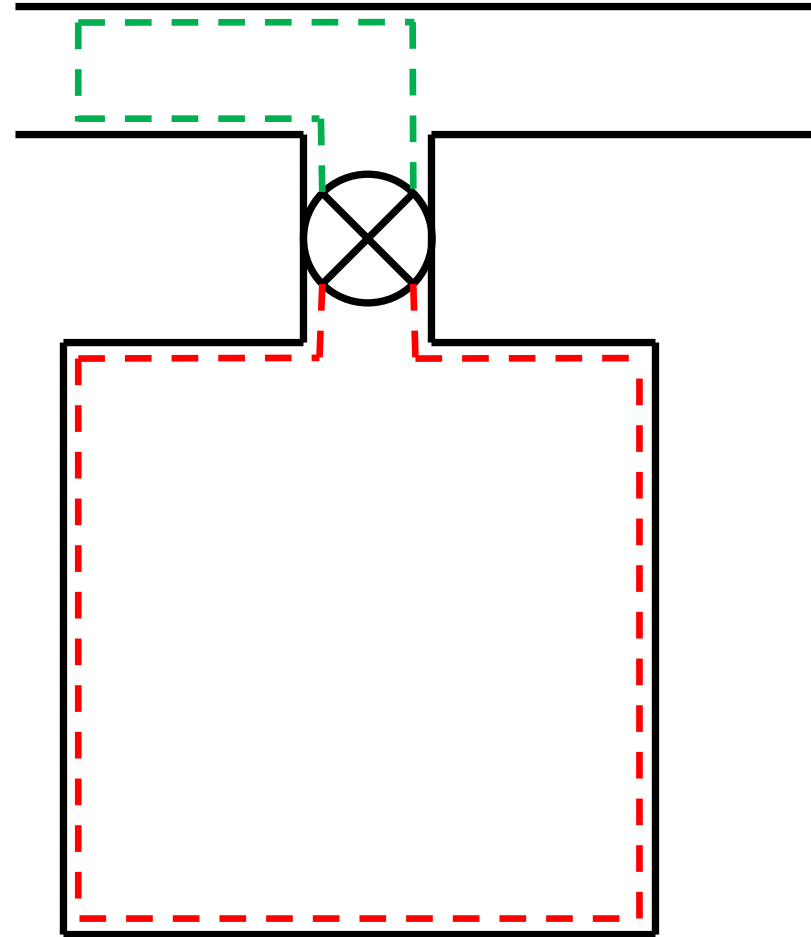


Fig.2. Closed System

Discharging Process

From physical viewpoint the discharge of a fluid from a pressurised vessel is similar to the preceding case of charging of a vessel. However, from thermodynamic point of view there is little similarity.

Now we should apply the conservation of the energy principle to the discharging process. We shall again neglect the kinetic and potential energies associated with either the control volume or the mass leaving the control volume.

$$\cancel{\delta Q} - \cancel{\delta W} + \sum_{in} \left(\cancel{h} + \cancel{\frac{V^2}{2}} + \cancel{gz} \right) \cancel{dm_i} - \sum_{out} \left(\cancel{h} + \cancel{\frac{V^2}{2}} + \cancel{gz} \right) \cancel{dm_e} = dU_{CV}$$

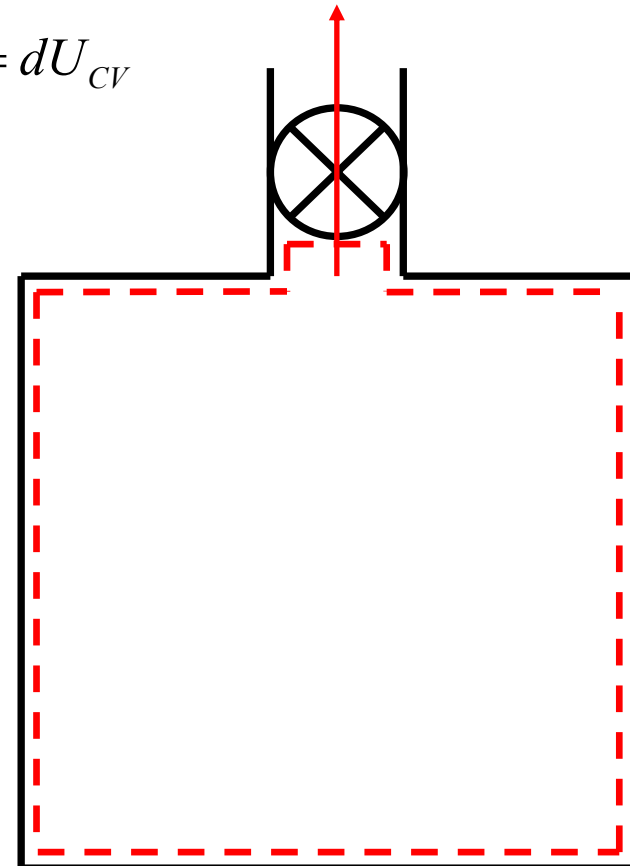
$$hdm = dU_{CV}$$

$$hdm = d(mu) = mdu + udm$$

This problem is unlike the charging of the vessel from a pressurised line in that the properties of the fluid crossing the control surface are continually changing. For this we have to write the conservation equation in its differential form.

$$(h - u)dm = mdu$$

$$\frac{du}{(h - u)} = \frac{dm}{m}$$



It should be noted here that the specification of the property values such as h , u for mass within the control volume requires that the fluid be essentially in equilibrium at all times. Hence the process of discharging the gas is quasistatic and one of uniform state.

The above equation is **completely general** i.e. we have not specified the nature of the fluid leaving the pressurised tank. We may introduce two further relationships without losing this generality

Assuming the **volume of the control volume remains constant** with time then from the basic definition:

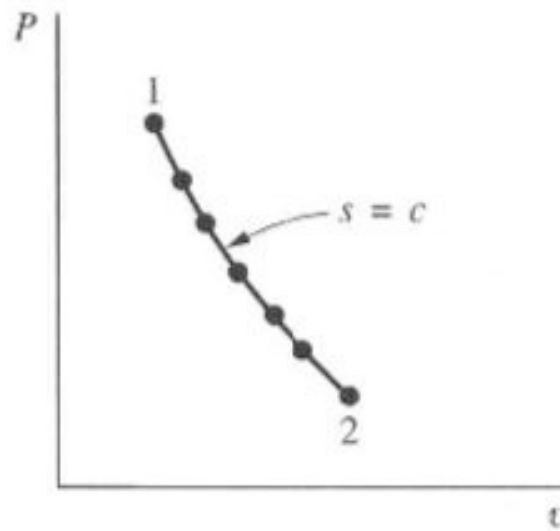
$$V = mv \Rightarrow dV = vdm + dv m = 0 \Rightarrow \frac{dm}{m} = -\frac{dv}{v}$$

Using the definition of enthalpy and above equation

$$\frac{du}{(h - u)} = \frac{dm}{m} \Rightarrow \frac{du}{Pv} = -\frac{dv}{v} \quad \text{or } du + Pdv = 0$$

Comparison of the above equation with the 1st Tds relation ($Tds = du + pdv$) for a simple compressible substance concludes that during the flow from a pressurised tank under the stated assumption or idealization the specific entropy within the control volume remains constant. The total entropy of the control volume decreases, since the amount of mass decreases. The overall process is highly irreversible. So that the net entropy change of control volume plus its environment must be positive.

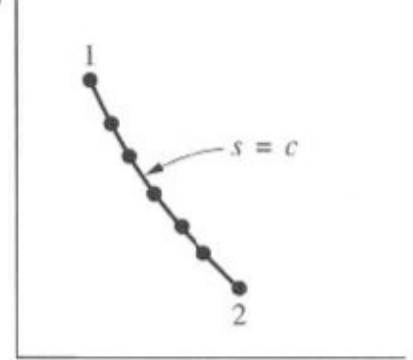
Figure illustrates the process path for the discharging of a fluid from a pressurised tank. The path is for the fluid remains inside the tank. Note that such process must fulfil the following assumption



- ❖ Adiabatic and rigid control volume
- ❖ Equilibrium within the tank at any instant (Uniform state)
- ❖ Negligible kinetic and potential energy changes

The fact that the process is isentropic is not surprising, since it is adiabatic and quasistatic and all internal dissipative effect and finite gradients have been ruled out. That is the process within the tank is adiabatic and internally reversible.

Special Example: Flow of an ideal gas from a pressurised tank. Assume the gas to be calorically perfect gas i.e. constant specific heats



A relation for mass remaining in the tank at any instant can be obtained. We noted earlier for a constant volume (total volume) process that

$$\frac{dm}{m} = -\frac{dv}{v}$$

For calorically perfect gas the isentropic relation is given as:

There are many other type of transient flow problems of theoretical and practical importance. The solution of each type depends on the restrictions or idealisations placed over the process. In every case it is best to begin with basic or fundamental equations and to proceed logically from that point

$$Tv^{\gamma-1} = c$$

$$\ln T + (\gamma - 1) \ln v = 0$$

$$\frac{dT}{T} = (1 - \gamma) \frac{dv}{v} = (\gamma - 1) \frac{dm}{m}$$

$$\ln T^{1/(\gamma-1)} = \ln m$$

$$\Rightarrow \left(\frac{T_2}{T_1} \right)^{1/(\gamma-1)} = \left(\frac{m_2}{m_1} \right) = \left(\frac{P_2}{P_1} \right)^{1/\gamma}$$

$$h_{avg} = \frac{1}{m_{out}} \int_1^2 h dm$$

From conservation of mass $m_{out} = (m_1 - m_2)_{tank}$

Approximating h_{avg} as $[(h_1 + h_2)/2]_{tank}$

$$Q - \int h dm = \Delta U_{CV}$$

$$Q - \frac{h_1(m_1 - m_2)}{2} - \frac{h_2(m_1 - m_2)}{2} = m_2 u_2 - m_1 u_1$$

$Q \approx 0$ the tank is insulated

$$m_1 \gg m_2 \approx 0$$

$$h_2 = 2u_1 - h_1$$

Emptying a pressurised incompressible fluid: $u = cT$ $h = cT + vp$

$$T_2 = T_1 - (v/c)(p_1 + p_2)$$

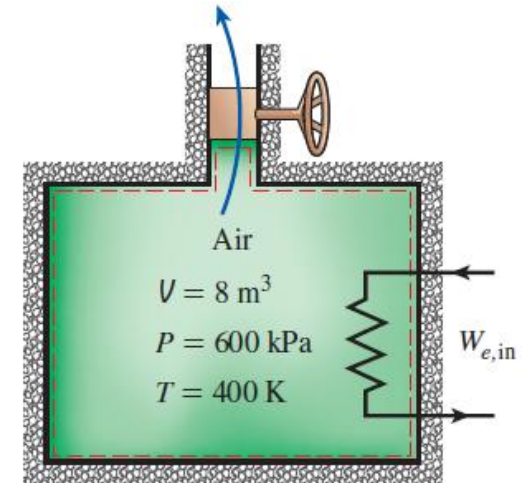
Emptying a pressurised ideal gas: $u = c_v T$ $h = c_p T$

$$T_2 = T_1 \left((2/\gamma) - 1 \right)$$

An insulated **8 m³ rigid tank** contains air at **600 kPa and 400 °C**. A valve connected to the tank is now opened, and air is allowed to escape until the pressure inside drops to **200 kPa**. The air temperature during the process is kept constant by electric resistance heater placed in the tank. Determine the electric energy supplied to the air during the process

Assumptions:

1. Unsteady process
2. The kinetic potential energies are neglected
3. The tank is insulated so the heat interaction is zero
4. Air is an ideal gas with variable specific heat



From conservation of mass $m_{out} = (m_1 - m_2)_{tank}$

$$W_{e,in} - m_{out}h_{out} = m_2u_2 - m_1u_1$$

$$m_1 = \frac{P_1V_1}{RT_1} = \frac{600 \times 8}{0.287 \times 400} = 41.81 \quad m_2 = \frac{P_2V_2}{RT_2} = \frac{200 \times 8}{0.287 \times 400} = 13.94$$

$$m_{out} = m_1 - m_2 = 41.81 - 13.94 = 27.87$$

From the table the final and initial internal energy of the air: $u_2 = u_1 = 286.16 \text{ kJ/kg}$

Enthalpy of the air leaving the control volume : $h_{out@400K} = 400.98 \text{ kJ/kg}$

$$\begin{aligned} W_{e,in} &= m_{out}h_{out} + m_2u_2 - m_1u_1 \\ &= 27.87 \times 400.98 + 13.94 \times 286.16 - 41.81 \times 286.16 = 3200 \text{ kJ} \end{aligned}$$

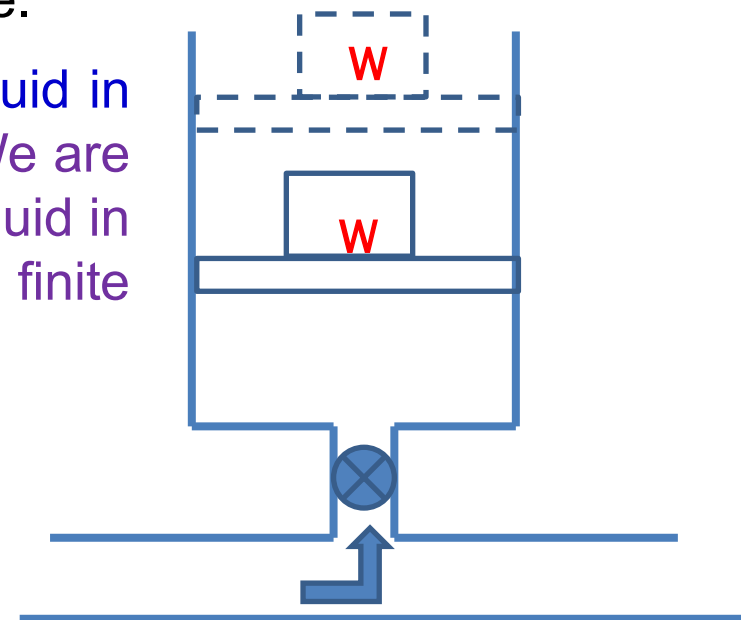
Transient Analysis with Boundary Work

Consider the situation shown in the figure. A weighted piston cylinder assembly initially containing a known quantity of gas at a known equilibrium state is connected through a valve to a high pressure line.

To simplify the analysis it is assumed that the fluid in the line is same as that in the piston cylinder. We are interested in final temperature and mass of the fluid in the cylinder after the volume has changed by finite amount

As the upper boundary will move, the control volume shown in the figure change size and boundary work occurs

Note: Even though the pressure in the line may be substantially higher than the pressure within the cylinder maintained by the weighted piston, we must assume that the entering gas quickly equilibrates with the gas already present in the cylinder. Thus a uniform and constant pressure is maintained by the piston on the gas. The work done by the gas on the surrounding:



$$W_{boundary} = \int P dV = P(V_f - V_i)$$

We shall assume the process is adiabatic, entering velocity is negligible. Although the centre of mass of the control volume changes in elevation, this potential energy effect is negligible. As a result $e=u$

$$Q - W_{Shaft} - W_{boundary} + \int (u + Pv) dm = \Delta U_{CV}$$

$$-W_{boundary} + \int h dm = \Delta U_{CV}$$

$$-P(V_f - V_i) + \int h dm = m_f u_f - m_i u_i$$

As the flow through the pipeline is steady, we can integrate $\int h dm$ directly

$$-P(V_f - V_i) + h_L m_L = m_f u_f - m_i u_i$$

$$m_L = m_f - m_i$$

Thus we have one equation but two unknowns m_f and u_f

The 2nd required relationship in this is found from a knowledge of the final volume

$$\text{Note: } V_f = m_f v_f$$

A piston cylinder assembly is attached through a valve to a source of air which is flowing in steady state through a pipe. Initially , inside the cylinder the volume is **0.01 m³**. The temperature of the air inside is **40 °C** and the pressure is **1 bar**. The valve is then opened to the air supply line which is maintained at **6 bar** and **100 °C**. If the cylinder pressure remains constant, what will be the final equilibrium temperature, and how much mass will have entered through the valve, when the cylinder volume reaches **0.02 m³**

Assumption: **The gas is ideal with constant specific heat**

$$-P(V_f - V_i) + h_L(m_f - m_i) = m_f c_v T_f - m_i c_v T_i$$

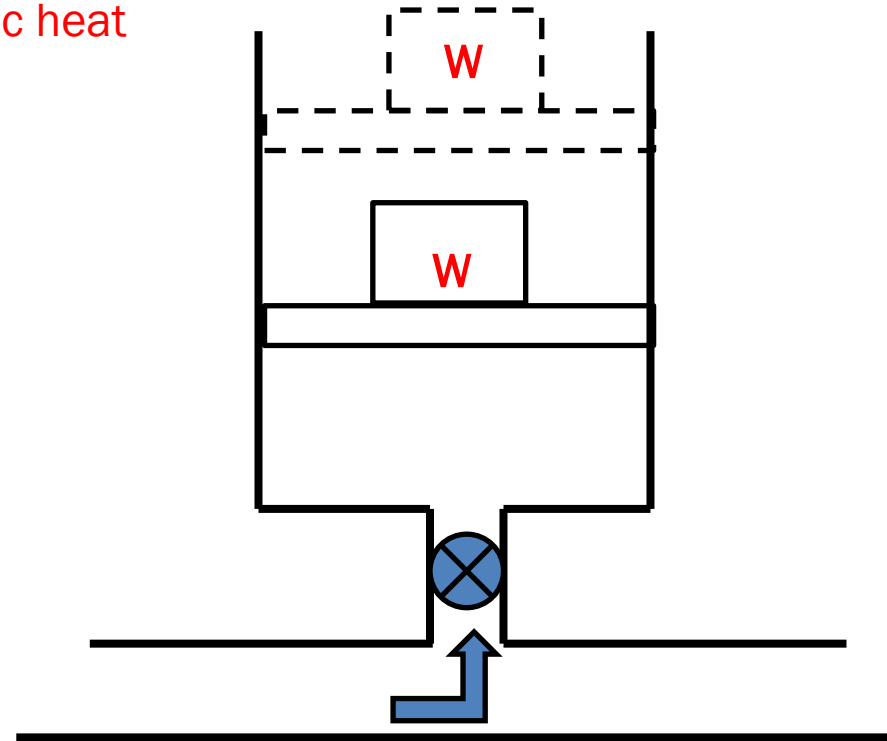
$$= m_f c_v \frac{P_f V_f}{m_f R} - m_i c_v T_i$$

$$= \frac{P V_f}{\gamma - 1} - m_i c_v T_i$$

$$m_i = \frac{P V_i}{R T_i} = \frac{100 \times 0.01}{0.287 \times 313} = 0.0111 \text{ [kg]}$$

$$\begin{aligned} & -P(V_f - V_i) + c_p T_L(m_f - m_i) \\ &= \frac{P V_f}{\gamma - 1} - m_i c_v T_i \end{aligned}$$

$$\begin{aligned} & -1 \times 10^5(0.02 - 0.01) + 1001 \times 373(m_f - 0.0111) \\ &= \frac{1 \times 10^5 \times 0.02}{1.4 - 1} - 0.0111 \times 728 \times 313 \Rightarrow m_f = 0.0204 \text{ [kg]} \end{aligned}$$



Thus the mass entering the assembly during the filling process is:

$$m_L = m_f - m_i = 0.0204 - 0.0111 = 9.3[\text{g}]$$

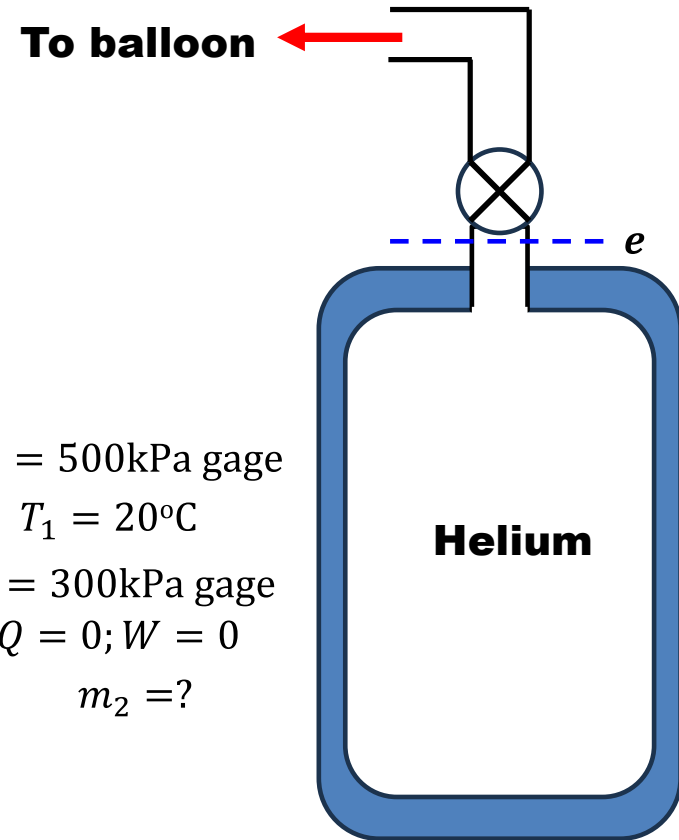
$$T_f = \frac{PV_f}{Rm_f} = \frac{100 \times 0.02}{0.287 \times 0.0204} = 68.6 \text{ } [^{\circ}\text{C}]$$

Thus the temperature of the resulting mixture in the assembly increases nearly 30 $[^{\circ}\text{C}]$ above the initial temperature

If the fluid had been a real gas, the solution would not have been quite as straight forward. In this later case the property relations are usually not known in an explicit algebraic form. Consequently, an iterative process often is required during the numerical evaluation.

A balloon is to be partially filled at atmospheric pressure with helium from a well insulated storage tank that has a volume of 6.20 m^3 . At the time of filling atmospheric pressure is 95.0 kPa . The helium in the tank is initially in the gauge pressure of 500 kPa and 20°C before the valve on the tank is opened. Assuming that conditions are uniform throughout the tank at any instant and heat transfer is negligible, determine the mass of helium that has been put into the balloon when the pressure gauge in the storage tank reads 300 kPa . For helium under these conditions, $h=1.667u=5.193T$, with h and u in kJ/kg and T in K

Solution: The mass of helium that has been put into the balloon is the difference the masses initially and finally in the storage tank. The initial mass in the tank can be calculated from ideal-gas equation of state., since the pressure temperature and volume are known.



$$P_1 = 500 \text{ kPa gage}$$

$$T_1 = 20^\circ\text{C}$$

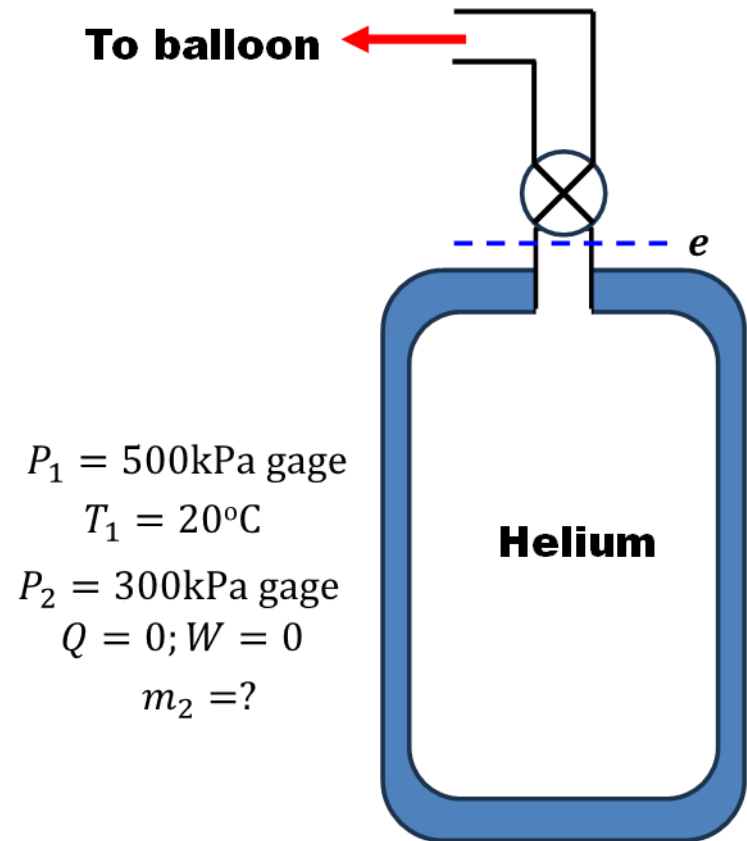
$$P_2 = 300 \text{ kPa gage}$$

$$Q = 0; W = 0$$

$$m_2 = ?$$

We can not calculate the final mass in the tank in this manner, because the final temperature is unknown, so we have two unknowns m_2 and T_2 in the ideal gas equation of state. We see in the problem statement that T is related to u , and another relationship involving m_2 , T_2 , and u_2 is the 1st law applied to the open system defined as the Helium within the tank at any instant.

Since e where the helium leaves the system is just upstream of the valve so the kinetic energy at e is very small. Also the properties of helium at e at any instant are equal to the properties of helium inside the tank. Since the properties of helium within the tank varies with time, we will use a differential form of the 1st law.



Calculation: As an infinitesimal mass of helium δm_e flows from the system, the mass of the helium within the system changes by an amount dm and $\delta m = -dm$

Since $Q=0$ and $W=0$

$$\left(\begin{array}{c} \text{Decrease in stored energy} \\ \text{of helium tank} \end{array} \right) = \left(\begin{array}{c} \text{stored energy} \\ \text{of } \delta m \text{ leaving tank} \end{array} \right) + \left(\begin{array}{c} \text{Flow work} \\ \text{of } \delta m \text{ leaving tank} \end{array} \right)$$

$$-dU = e\delta m + pv\delta m = (e + pv)\delta m_e$$

As noted above, section e is just up stream of the small valve opening, so the kinetic energy at the exit is negligibly small. Potential energy change can be neglected. then

$$e_e = u_e$$

$$-dU = (u_e + p_e v_e)\delta m_e$$

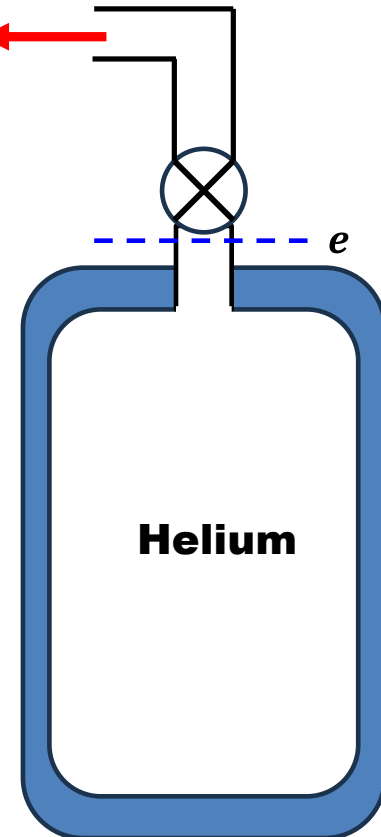
$$P_1 = 500\text{kPa gage}$$

$$T_1 = 20^\circ\text{C}$$

$$P_2 = 300\text{kPa gage}$$

$$Q = 0; W = 0$$

$$m_2 = ?$$



Since the properties of the helium leaving are the same as those of the helium in the tank.

$$u_e = u; p_e = p; v_e = v$$

Recalling also that: $\delta m = -dm$

$$-dU = -(u + pv)\delta m$$

$$d(mu) = (u + pv)dm$$

$$\Rightarrow mdu + udm = (u + pv)dm$$

$$\Rightarrow mdu = pvd m$$

Since internal energy can be expressed in terms of T as

$$1.667u = 5.193T \text{ or } u = 3.115T, \text{ so that } du = 3.115dT$$

The energy balance becomes: $3.115mdT = pvd m$

From ideal gas equation of state, $T = \frac{pv}{R}$

So making the substitution: $3.115mdT = pvd m$

$$\begin{aligned} \frac{3.115m}{R} d(pv) &= pvd m \\ \Rightarrow \frac{dm}{m} &= \frac{3.115}{R} \frac{d(pv)}{pv} \\ \Rightarrow \ln\left(\frac{m_2}{m_1}\right) &= \frac{3.115}{R} \ln\left(\frac{p_2 v_2}{p_1 v_1}\right) \\ \Rightarrow \left(\frac{m_2}{m_1}\right) &= \left(\frac{p_2 v_2}{p_1 v_1}\right)^{\frac{3.115}{R}} = \left(\frac{p_2 V_2 m_1}{p_1 V_1 m_2}\right)^{\frac{3.115}{R}} \end{aligned}$$

As we know that the tank volume is constant, $V_2 = V_1 = V$,

$$\left(\frac{m_2}{m_1}\right) = \left(\frac{p_2}{p_1}\right)^{\frac{1.5}{2.5}} = \left(\frac{p_2}{p_1}\right)^{0.6}$$
$$\Rightarrow m_2 = m_1 \left(\frac{p_2}{p_1}\right)^{0.6}$$

The mass of helium that has been put into the balloon equals the decrease in the mass of helium in the storage tank:

$$m_{to\ balloon} = m_1 - m_2 = m_1 - m_1 \left(\frac{p_2}{p_1}\right)^{0.6} = m_1 \left[1 - \left(\frac{p_2}{p_1}\right)^{0.6}\right]$$

Evaluating m_1 by the ideal gas equation of state, and recalling that absolute pressure is the sum of gage pressure and atmospheric pressure we have

$$m_{to\ balloon} = \frac{p_1 V}{RT_1} \left[1 - \left(\frac{p_2}{p_1}\right)^{0.6}\right]$$
$$= \frac{595(6.20)}{2.077(293)} \left[1 - \left(\frac{395}{595}\right)^{0.6}\right] = 1.32 \text{ [kg]}$$

A 4-L pressure cooker has an operating pressure of 175 kPa. Initially, one-half of the volume is filled with liquid and the other half with vapor. If it is desired that the pressure cooker not run out of liquid water for 75 min, determine the highest rate of heat transfer allowed.

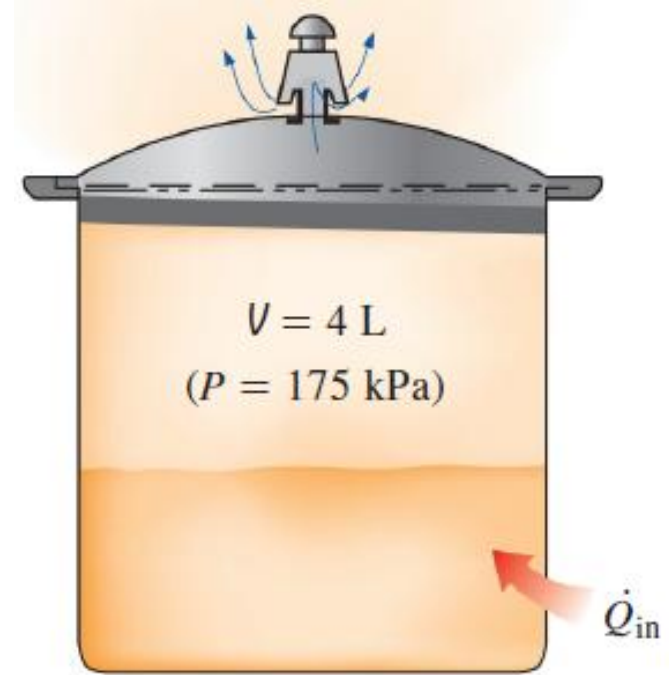
$$Q - W + \int_i \left(h + \frac{V^2}{2} + dz \right) dm_i - \int_e \left(h + \frac{V^2}{2} + dz \right) dm_e = \Delta U_{CV}$$

$$\dot{Q}_{in} \Delta t - h_e \Delta m_e = \Delta U_{CV}$$

$$\dot{Q}_{in} \Delta t - h_{g@175kPa} \Delta m_e = \Delta U_{CV}$$

$$\Delta m_e = m_i - m_f = \frac{V_{L,in}}{v_f} + \frac{V_{g,in}}{v_g} - \frac{V_{total}}{v_g}$$

$$\Delta U_{CV} = m_f u_f - m_i u_i = \left(\frac{V_{total}}{v_g} \right) u_g - \left(\frac{V_{L,in}}{v_f} \right) u_f - \left(\frac{V_{g,in}}{v_g} \right) u_g$$



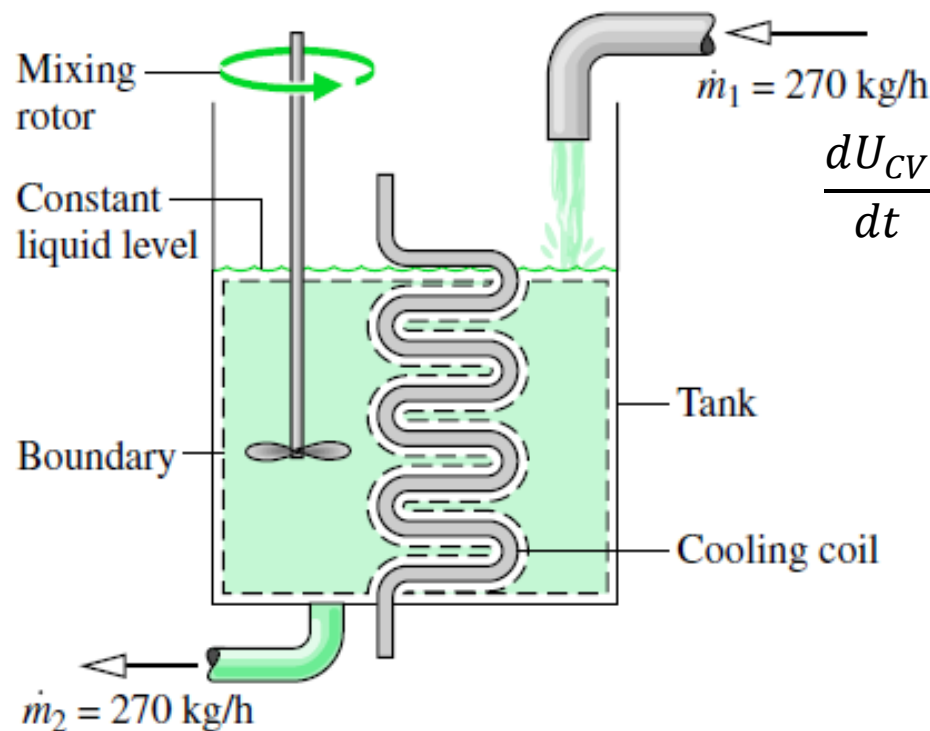
At 175 kPa

$$v_f = 0.001075 \text{ m}^3/\text{kg} \quad u_f = 486.82 \text{ kJ/kg} \quad h_f = 487.01 \text{ kJ/kg}$$

$$v_g = 1.0073 \text{ m}^3/\text{kg} \quad u_g = 2524.5 \text{ kJ/kg} \quad h_g = 2700.2 \text{ kJ/kg}$$

$$\Delta m_e = (2 \times 10^{-3} / v_g) + (2 \times 10^{-3} / v_f) - (4 \times 10^{-3} / v_g)$$

A tank containing 45 kg of liquid water initially at 45°C has one inlet and one exit with equal mass flow rates. Liquid water enters at 45°C and mass flow rate of 270 kg/h. A cooling coil immersed in the water removes energy at a rate of 7.6 kW. The water is well mixed by a paddle wheel so that the water temperature is uniform throughout. The power input to the water from the paddle wheel is 0.6 kW. The pressure at inlet and exit are equal and all kinetic and potential energy effects can be ignored. Plot the variation of water temperature with time



$$\frac{dU_{CV}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2)$$

$$\frac{dU_{CV}}{dt} = \frac{d(m_{cv}u)}{dt} = m_{cv} \frac{du}{dt} + u \frac{dm_{cv}}{dt} = m_{cv} \frac{du}{dt}$$

$$\frac{du}{dt} = \frac{du}{dT} \frac{dT}{dt} = c \frac{dT}{dt}$$

$$\frac{dU_{CV}}{dt} = m_{cv}c \frac{dT}{dt}$$

$$(h_1 - h_2) = c(T_1 - T_2) + v(P_1 - P_2)$$

$$\frac{dU_{CV}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2)$$

$$m_{cv}c \frac{dT}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}c(T_1 - T)$$

$$T = C_1 \exp\left(-\frac{\dot{m}}{m_{cv}}t\right) + \left(\frac{\dot{Q}_{cv} - \dot{W}_{cv}}{\dot{m}c}\right) + T_1$$

$$\text{At } t = 0 \quad T = T_1 \quad T = T_1 + \left(\frac{\dot{Q}_{cv} - \dot{W}_{cv}}{\dot{m}c}\right) \left[1 - \exp\left(-\frac{\dot{m}}{m_{cv}}t\right)\right]$$